

Donghwa Kim

Fullstack Engineer (Realtime / Media Systems)

Fullstack engineer with 4+ years of experience building production systems across frontend, backend, and infrastructure.

Specialized in real-time communication (WebRTC), multimedia processing (FFmpeg), and 3D visualization (Three.js). Strong track record of delivering production-ready systems under real-world constraints, including hardware integration, low-latency communication, and performance-critical environments.

Focused on pragmatic system design, fast iteration, and turning prototypes into scalable, revenue-generating products.

- Email: devmemorydh@gmail.com
 - GitHub: <https://github.com/devmemory>
 - Portfolio: <https://devmemory.github.io/#!/portfolio>
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TECHNICAL SKILLS

Frontend

React, Next.js, TypeScript, Zustand, Tailwind

Backend

Node.js, NestJS, PostgreSQL, Redis, TypeORM

Infrastructure / DevOps

AWS (EC2, S3, Route53), Docker, NGINX, GitHub Actions

Realtime / Media Systems

WebRTC, WebSocket, HLS, FFmpeg

EXPERIENCE

Outloud Company — Lead Developer

Aug 2025 – Jan 2026

Led end-to-end development of an offline music booth kiosk platform integrating hardware devices and real-time media processing.

- Re-architected a prototype system into a production-ready platform deployed in physical stores
- Designed Electron-based architecture combining browser UI with native hardware control

- Built media processing pipeline using FFmpeg and HLS streaming
- Integrated external devices (payment terminals, cameras, photo printers)
- Established CI/CD pipelines with AWS, Docker, and GitHub Actions
- Built operational tooling using Webhooks and Google Sheets for rapid issue handling

Impact

- Delivered MVP in 4 weeks, full production system in 4 months
 - Successfully launched in real stores and contributed to revenue generation
 - Improved deployment stability and reduced incident response time
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Luxrobo — Frontend Engineer

Mar 2024 – Apr 2025

Worked on AI circuit design and smart parking systems under tight deadlines.

- Led frontend architecture design and implementation
- Built 3D PCB viewer using Three.js and circuit editor using JointJS
- Developed internal tooling for GraphQL code generation, improving dev efficiency
- Implemented real-time CCTV emergency system using WebRTC
- Set up CI/CD pipelines with Docker, NGINX, and GitHub Actions

Impact

- Delivered investor-facing prototype for AI circuit design platform
 - Reduced delivery timeline by 2 weeks for parking system → enabled faster go-to-market
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Gaon Platform — Frontend Engineer

Mar 2023 – Sep 2023

Improved and stabilized an AI-based risk management system under deadline pressure.

- Refactored frontend architecture and introduced modular structure
- Migrated polling-based system to real-time socket communication
- Designed authentication and authorization system (JWT + role-based access)
- Built data visualization with D3.js and Highcharts

Impact

- Met delivery deadlines despite initial delays
- Improved system performance, stability, and data visualization quality

Additional Project (2 weeks)

- Independently built pump failure diagnostic visualization tool
 - Delivered 1 week ahead of schedule
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Lexynaptics — Frontend Engineer

Jun 2022 – Dec 2022

Built MVP for legal consultation platform targeting investor validation.

- Designed frontend architecture for web and mobile
- Implemented 1:1 real-time video consultation using WebRTC
- Integrated payments, identity verification, and third-party APIs
- Built user analytics/event tracking system

Impact

- Delivered MVP before investor meetings
 - Enabled successful investment preparation
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Reindeer — App Developer

Apr 2021 – Jun 2022

Worked on travel guide and reservation mobile app.

- Refactored legacy codebase and improved performance
- Built custom map solution due to lack of suitable SDK
- Implemented payment and external API integrations
- Improved UI/UX and adapted new design system

Impact

- Improved performance by 10+ FPS
 - Reduced crash rate by ~90%
 - Stabilized service for continued operation and expansion
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PROJECTS

PCB 3D Viewer

<https://devmemory.github.io/gerber-3d-viewer>

- Built 3D visualization tool for PCB Gerber files using Three.js
 - Implemented CI/CD pipeline for automated deployment
 - Received collaboration proposal from PCBWay
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Cursor Ubuntu Installer

<https://github.com/devmemory/cursor-ubuntu-installer>

- Built and distributed shell installer for Cursor IDE on Ubuntu
 - Implemented automated version update system
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KakaoMap Flutter Package

https://pub.dev/packages/kakaomap_webview

- Developed and published Flutter package for Kakao Map integration
 - Used in 100+ applications
 - Achieved 27 likes on Pub.dev
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EDUCATION

Kongju National University

B.S. in Electrical, Electronics & Control Engineering

(2010 – 2017, including military service)